Assignment day - 5

1. Reverse a list in Python.

```
[1]: list1 = [100, 200, 300, 400, 500]
     list1.reverse()
     print(list1)
```

```
[500, 400, 300, 200, 100]
```

2. Write a program to add two lists index wise. Create a new list that contains the 0th index item from both the list, then the 1st index and so on till the last element. Any leftover items will get added at the end of the new list.

```
[2]: list1 = ["M", "na", "i", "Ke"]
     list2 = ["y", "me", "s", "lly"]
     result = [a + b for a, b in zip(list1, list2)]
     print(result)
```

```
['My', 'name', 'is', 'Kelly']
```

3. Given a list of no.s . Write a program to turn every item of a list into its sq.

```
[3]: numbers = [1, 2, 3, 4, 5, 6, 7]
     squared = [num ** 2 for num in numbers]
     print(squared)
```

```
[1, 4, 9, 16, 25, 36, 49]
```

4. Given a two-python list. Write a program to iterate both lists simultaneously and display items from list-1 in original order and items from list-2 in reverse order.

```
[4]: list1 = [10, 20, 30, 40]
     list2 = [100, 200, 300, 400]
     for a, b in zip(list1, reversed(list2)):
         print(a, b)
```

```
10 400
20 300
30 200
40 100
```

5. Write a program to ad item 7000 after 6000 in the following Python List.

```
[5]: list1 = [10, 20, [300, 400, [5000, 6000], 500], 30, 40]
     list1.append(7000)
     print(list1)
```

```
[10, 20, [300, 400, [5000, 6000], 500], 30, 40, 7000]
```

6. You have given a nested list. Write a program to extend it by adding the sub list [“h”, “l”, “j”] in such a way that it will look like the following list.

```
list1 = ["a", "b", ["c", ["d", "e", ["f", "g"], "k"], "l"], "m", "n"]
sub_list = ["h", "i", "j"]

list1[2][1][2].extend(sub_list)

print(list1)
```

```
['a', 'b', ['c', ['d', 'e', ['f', 'g', 'h', 'i', 'j'], 'k'], 'l'], 'm', 'n']
```

7. You have given a Python List. Write a program to find value 20 in the list and if it is present, replace it with 200. Only update the first occurrence of an item.

```
list1 = [5, 10, 15, 20, 25, 50, 20]
index = list1.index(20)
list1[index] = 200
print(list1)
```

```
[5, 10, 15, 200, 25, 50, 20]
```

8. Given a Python list. Write a program to remove all occurrences of item 20.

```
list1 = [5, 20, 15, 20, 25, 50, 20]
result = [x for x in list1 if x != 20]
print(result)
```

```
[5, 15, 25, 50]
```

9. Generate a Python list of all the even no.s b/w 4 to 30.

```
even_numbers = [x for x in range(4, 30) if x % 2 == 0]
print(even_numbers)
```

```
[4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28]
```

10. Find the largest item from the given list.

```
x = [4, 6, 8, 24, 12, 2]
print(max(x))
```

```
24
```

11. Use a loop to display elements from a given list present at odd index positions.

```
list1 = [10, 20, 30, 40, 50]
for i in range(1, len(list1), 2):
    print(list1[i])
```

```
20
40
```

12. Write a program to count the total no. of digits in a no. using a while loop.

```
num = 75869
count = 0
while num != 0:
    num //= 10
    count += 1
print(count)
```

```
5
```

13. Print list in reverse order using loop.

```
list1 = [10, 20, 30, 40, 50]
for i in range(len(list1)-1, -1, -1):
    print(list1[i])
```

```
50
40
30
20
10
```

14. Write a program to display only those no.s from a list that satisfy the following conditions.

→The no. must be divisible by 5

→If the no. is greater than 150 then, skip it and move to the next no.

→If the no. is greater than 500 then, stop the loop.

```
numbers = [12, 75, 150, 180, 145, 525, 50]
for num in numbers:
    if num > 500:
        break
    if num > 150:
        continue
    if num % 5 == 0:
        print(num)
```

```
75
150
145
```

15. Convert two lists in dictionary. Write a Python program to convert them into a dictionary in a way that item from list-1 is the key and item from list-2 is the valve.

```
keys = ['Ten', 'Twenty', 'Thirty']
values = [10, 20, 30]
result = dict(zip(keys, values))
print(result)
```

```
{'Ten': 10, 'Twenty': 20, 'Thirty': 30}
```

16. Merge the two python dictionaries into one.

```
dict1 = {'Ten': 10, 'Twenty': 20, 'Thirty': 30}
dict2 = {'Thirty': 30, 'Fourty': 40, 'Fifty': 50}
dict1.update(dict2)
print(dict1)
```

```
{'Ten': 10, 'Twenty': 20, 'Thirty': 30, 'Fourty': 40, 'Fifty': 50}
```

17. Print the value of key ‘history’ from the below dictionary.

```
sampleDict = {
    "class": {
        "student": {
            "name": "Mike",
            "marks": {
                "physics": 70,
                "history": 80
            }
        }
    }
}
print(sampleDict['class']['student']['marks']['history'])
```

```
80
```

18. Initialise dictionary with the default values. In python, we can initialise the keys with the same values.

```
employees = ['Kelly', 'Emma']
defaults = {"designation": 'Developer', "salary": 8000}
result = dict.fromkeys(employees, defaults)
print(result)
```

```
{'Kelly': {'designation': 'Developer', 'salary': 8000}, 'Emma': {'designation': 'Developer', 'salary': 8000}}
```

19. Write a python program to create a new dictionary by extracting the mentioned keys from the below dictionary.

```
sample_dict = {
    "name": "Kelly",
    "age": 25,
    "salary": 8000,
    "city": "New york"
}
keys = ["name", "salary"]
new_dict = {k: sample_dict[k] for k in keys}
print(new_dict)
```

```
{'name': 'Kelly', 'salary': 8000}
```

20. Delete a list of keys from a dictionary.

```
sample_dict = {
    "name": "Kelly",
    "age": 25,
    "salary": 8000,
    "city": "New york"
}
keys = ["name", "salary"]
for k in keys:
    sample_dict.pop(k)
print(sample_dict)
```

```
{'age': 25, 'city': 'New york'}
```

21. We know how to check if the key exists in a dictionary. Sometimes it is required to check if the given value is present. Write a python program to check if the value 200 exists in the following dictionary.

```
sample_dict = {'a': 100, 'b': 200, 'c': 300}
if 200 in sample_dict.values():
    print("200 present in a dict")
```

```
200 present in a dict
```

22. Write a program to rename a key city to a location in the following dictionary.

```
sample_dict = {
    "name": "Kelly",
    "age": 25,
    "salary": 8000,
    "city": "New york"
}
sample_dict['location'] = sample_dict.pop('city')
print(sample_dict)
```

```
{'name': 'Kelly', 'age': 25, 'salary': 8000, 'location': 'New york'}
```

23. Get the key of a min. value from the following dictionary.

```
sample_dict = {
    'Physics': 82,
    'Math': 65,
    'history': 75
}
min_key = min(sample_dict, key=sample_dict.get)
print(min_key)
```

```
Math
```

24. Write a python program toc change Brad's salary to 8500 in the following dictionary.

```
sample_dict = {
    'emp1': {'name': 'Jhon', 'salary': 7500},
    'emp2': {'name': 'Emma', 'salary': 8000},
    'emp3': {'name': 'Brad', 'salary': 500}
}
sample_dict['emp3']['salary'] = 8500
print(sample_dict)
```

```
{'emp1': {'name': 'Jhon', 'salary': 7500}, 'emp2': {'name': 'Emma', 'salary': 8000}, 'emp3': {'name': 'Brad', 'salary': 8500}}
```